

SPECTRAVIDEO USERS GROUP

WELLINGTON

NEWSLETTER

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Meetings are held on the third Monday of the month at 7.30 pm in the Staff Room of Newlands Intermediate, Bracken Road, Newlands, Wellington. Attendance currently averages 18-15 with ideas, information, problems and matters of interest being shared.

COMMITTEE MEMBERS

Chairman	Jim Oliver	ph 665 285
Secretary/Treasurer	Ross Fletcher	ph 788 454
Librarian	Scotty Baxter	ph 784 619

Public Domain BASIC	Jim Oliver
Public Domain CP/M	Ross Fletcher

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WELLINGTON

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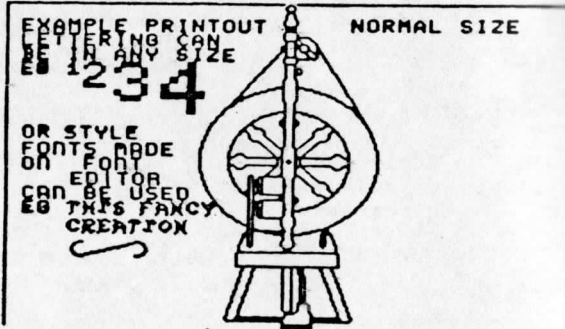
From your Editor,

Well here we are again! With a little help from Steve Hanson I finally got the last newsletter published and posted, albeit a little late to some of you.

There have been a few glitches in bringing the membership files up to date - Steve had to do a rebuild on the database before it could be used again. If you know of anyone not getting their Newsletter who ought to be then please let me know pronto. Also if you are due some backcopies then yell.

There is some pretty good content this issue and more articles to come in the next - for instance, how would you like to be able to produce printed graphics like this on your SVI...

This is produced by a 780 line Basic program submitted by Michael Warwick, Tokoroa. I am holding it over to the next issue to encourage people to renew their subscriptions - this and a number of other fascinating articles as well. In the meantime see Warwicks letter further on.



On the subject of subs - a lot of the recipients of this issue have expired - as members anyway. Please check your label and if you are overdue then lets have your \$18 or else you will miss out next time! To encourage would be contributors, I am offering an Editors Prize of \$20 for the best article over each of three consecutive issues beginning with this one. Judging is purely arbitrary with the Editor having sole discretion and no discussion being entered into!

The Editors Prize this issue goes to BOB CURWOOD for his 80 col PRINT screen program - in the mail Bob, thank you! Finally, for you Wordstar freaks, did you realise that ^G plus a cursor key produces the same result as the ^GE, ^GX, ^GB and ^GD operations?

.....John

CORRECTION

Last Issue's Cursor Program will not run as it is supposed to if you folowed the printed listing! You must add line

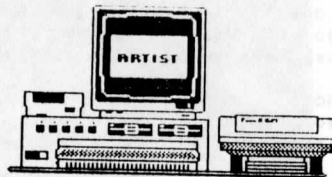
2045 RETURN

The loop is now complete and the program flow correct. It will now return and form a continous loop which will indeed show the changing cursor.

...courtesy Jim Oliver

LETTERS TO THE ED... (note my address is now published on the front cover - if you have a disk drive please send contributions by disk - I will return them ... John)

[Michael Warwick - 1 page]



TO THE ED

I STARTED THE ENCLOSED ART PROGRAM (WITH THE ORIGINAL TITLE OF 'ARTIST') SOME TIME AGO, AND SINCE THEN HAVE BEEN ADDING TO AND ALTERING IT. I WAS READING THE LAST ISSUE OF THE CLUB MAGAZINE, AND CAME ACROSS THE FEEDBACK WANTED ARTICLE, AND NOW I AM SENDING IN THIS PROGRAM IN REPLY TO JIM OLIVER'S REQUEST FOR A PROGRAM WHICH WOULD PRINT MODIFIED FONTS TO SCREEN 1 FOR PRINTING. THIS PROGRAM CAN DO JUST THAT! (THE ROUTINE IS IN LINES 480-570). AND THIS PROGRAM ALSO LETS YOU DRAW YOUR OWN PICTURES TO GO WITH IT. PRESS [3] TO ENTER TYPE MODE, THEN TYPE IN WHATEVER YOU WANT, THE PROGRAM WILL GO DOWN TO THE NEXT LINE WHEN YOU GET TO THE EDGE, AND THEN PRESS ENTER TO RETURN TO DRAWING. PRESSING [^P] WILL ENTER LETTER-HEAD, OR [2] FOR DOUBLE-SIZED PRINT-WARNING-IT TAKES DOUBLE TIME. THE PROGRAM IS SET UP FOR A PANASONIC KX-P1080 PRINTER. IT USES EPSON CODES, AND SO THE ART PROGRAM SHOULD WORK FOR MOST PRINTERS.

ALSO FROM THAT ISSUE, HERE IS MY SOLUTION TO THE BRAIN-TEASER:

```

10 CLS
20 LINEINPUT "WIDTH OF SHEET OF METAL (IN cm) >";A$;W=VAL(A$)
30 IF W=0 THEN GOTO 20 ELSE PRINT
40 FOR C=.05 TO W/2 STEP .05:REM
50 NA = ((W-(C*2))^2)*C:REM          CORRECT TO 1/2 mm
60 IF NA>OA THEN OA=NA:S=C:REM      AREA OF DRIP TRAY
70 NEXT C                          IS IT THE LARGEST SO FAR?
80 PRINT "THE SIDES SHOULD BE";S;"cm HIGH TO MAKE THE DRIP TRAY HOLD THE MOST
   VOLUME"
90 PRINT "THE VOLUME WOULD BE";OA;"sq cm APPROXIMATELY"
```

THE PROGRAM WILL WORK OUT THE HIEGHT THE SIDES SHOULD BE FOR MAXIMUM VOLUME.

I AM NOW WORKING ON A BETTER, EASIER TO USE, AND FASTER ART PROGRAM, AND HOPE TO BE SENDING IT TO YOU FOR PUBLIC DOMAIN SOMETIME IN THE NOT TO DISTANT FUTURE, BUT DON'T EXPECT MIRACLES.

YOURS IN SPECTRAVIDEO,

Michael Warwick

P.S. LINE 200 IN THE ART PROGRAM WILL NEED TO BE CHANGED, UNLESS THE PROGRAM IS GOING TO BE USED IN A MENU ENVIRONMENT, ALSO LINES 90 & 100 IF YOU DON'T HAVE A DISK DRIVE (JUST LEAVE OUT THE 'I:').

BRAIN STRAIN DEPARTMENT

Answer to Brain Teaser No 1.

We have had only one reply to the Brain Teaser posed in the last issue, (thankyou Michael!). The lack of response to the challenge is strange since these sort of things usually produce a wealth of ideas as to how any problems should be best tackled. See how you go with Brainteaser No. 2 below and lets have your solution - the best one to be published. Here is my solution to No 1.

No 1. centered around trying to work out the height of the sides so as to obtain the maximum volume. If you look at the diagram below you will see that the volume of the tray will be:

$$V = (40 - 2 * x) * (40 - 2 * x) * x$$

[so that's where I got lost ... ed]

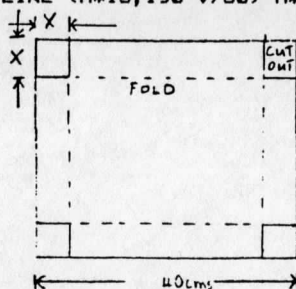
Since I said I could work to an accuracy of 0.5mm we assume that 0.5mm is the smallest height I would start with. We then increment the height from this and each time through the loop work out the new volume then compare it with the last result.

When we reach the maximum we have the answer printed out for all to see. I did it like this:

```
10 X=0.05: Y=0
20 V=(40-2*X)^2*X
30 IF V<Y THEN PRINT Y: END
40 Y=V: X=X+0.05: GOTO 20
```

Is that anything like your solution? A fancier solution can be obtained like this:

```
20 SCREEN 1: H=0.1
30 LINE (0,0)-(0,190),15: LINE (0,190)-(255,190),15
40 FOR X=0 TO 255 STEP 25.5: LINE (X,185)-(X,190),15: NEXT X
50 V=(40-2*X)^2*X
60 IF FL=1 THEN 70 ELSE IF V<MV THEN LINE (H*13,190-V/30)-H*13,
190),15: FL=1
70 MV=V
80 PSET (H*13,190-V/30),15
90 H=H+0.1
100 IF H>20 THEN 120
110 GOTO 50
120 GOTO 120
```



(Note: each division on the X axis is approximately 2)
... Jim Oliver

And for those wondering what the result was, the max volume could be obtained with sides of 6.67mm ...ed

BRAIN TEASER No 2.

In any well run Library you will see rows of Encyclopaedias neatly lined up on the shelves. However the other day I saw a 9 volume set (numbered from 1 to 9) which were shelved most untidily. There were 5 volumes on the top shelf and four volumes below these and they were all out of numerical order. The top 5 had Volume 3 in the middle.

I commented on the untidy arrangement and the assistant told me they were arranged specially like that for smart alecs. She said that the number formed by the Volume numbers on the top shelf was exactly twice the number formed by the numbers on the bottom shelf.

It took me more than a few minutes to work out a Basic program to calculate how the volumes were arranged. How about you?

... Jim Oliver

BOOK REVIEW

Hacker's Handbook III

Hugo Cornwall
Century Hutchinson Ltd
P.O. Box 40-086
AUCKLAND 10

I paid \$27.95 for my copy of this classic in its current updated 3rd edition, however most libraries should have at least one of the earlier editions available - if you haven't already scored it.

Wellington people can purchase it from Unity Bookshop, Willis Street.

My interest is in the area of communications, specifically setting up with a modem on the SVI and the two chapters, "Computer-to-Computer communications", and, "Hacker's Equipment" give some good basic instruction. If you want to go further there are chapters on Hacker's techniques, intelligence, and targets as well as on Networks, Videotex systems and intercepting Radio computer transmissions.

The book is well written and entertaining with exposes of some of the great "hacks", eg: the Prince Phillip/Prestel hack. If you have designs on the Wanganui Computer Centre you could do worse than read this book - particularly for its discussion on ethics of hacking and warnings about forbidden territories!

The appendices have good info on lots from RS232 to Modems. I am pleased to have it on my shelf and would recommend it to anyone looking to cut the best part of 30 bucks.

... ed

THE FOLLOWING ENABLES THE "PRINT" KEY
AS A HOT KEY TO DUMP AN 80
COLUMN SCREEN TO YOUR PRINTER.

```

10 |*****|
20 |*|
30 |* 80 Col: Page Print |
40 |* SVI Disk Basic |
50 |* Bob Curwood |
60 |* Christchurch Spectravideo |
70 |* Users Group |
80 |*|
90 |* Prints 23 lines of 80col |
100 |* screen. Change PAGE to |
110 |* 780H for 24 lines |
120 |*|
130 |*****|
140
150 |Version 1.0|
160
170 ORG 9000H
180 LOAD 9000H
190 CHSNS: EQU 03DCAH
200 CHGET: EQU 403DH
210 CHPLPT: EQU 3915H
220 BEEP: EQU 40BEH
230 HOOK: EQU 0FE79H
240 OFFSET: EQU 0F000H |Vidio ram start address
250 PAGE: EQU 730H |80 * 23 lines
260 |EQU 780H |80 * 24 lines
270
280 ALTERHK: LD HL,START
290 LD (HOOK+1),HL
300 LD A,0C3H
310 LD (HOOK),A
320 RET
330 START: LD A,8 |PRINT key pressed ?
340 OUT (96H),A
350 IN A,(99H)
360 AND 32
370 RET NZ |NO
380 PRNTON: IN A,(18) |Printer switched on ?
390 AND 1
400 RET NZ |NO
410 MAIN: LD HL,OFFSET
420 LD DE,OFFSET+PAGE
430 LD A,0AH
440 CALL CHPLPT |Clear printer buffer
450 LOOP: DI |Switch off interrupts
460 LD A,255
470 OUT (88),A |Switch in vidio ram
480 LD C,(HL)
490 INC A
500 OUT (88),A |Switch it out again
510 INC HL |UP the count
520 CTRLC: LD A,6 |1st test fot CTRL key
530 OUT (96H),A
540 IN A,(99H)
550 AND 2
560 JR NZ,NOINT |CTRL not pressed
570 LD A,2 |Now test for C key

```

```

580          OUT (96H),A
590          IN A,(99H)
600          AND 8
610          JR Z,EXIT          ;CTRL C is being pressed
620 NOINT:    LD A,C
630          CP 95              ;0 to 95 =space to tilde ~
640          JR C,ALPHA
650          CP 190             ;190 to 223 = inverse
660          JR NC,GRAPHIC
670 INVERSE:  ADD A,64          ;This gives italics on my
680 ;printer. Change to SBC A,63 for normal ASCII
690          JR PRTCHR
700 ALPHA:    ADD A,32
710          JR PRTCHR
720 GRAPHIC:  NOP              ;Watch this space!
730          NOP              ; (Next version)
740 PRTCHR:   CALL CHPLPT      ;Char in A out to printer
750          RST ZOH           ;HL = DE yet ?
760          JR NZ,LOOP        ;No
770          LD A,0AH
780          CALL CHPLPT        ;Print last line
790 EXIT:     CALL BEEP        ;Indicate printout finished
800          RET              ; or aborted
810          END
820
830

```

840 This routine is executed at the start of each VDP interrupt
850 (via the hook) and 1st checks if the PRINT key is being
860 pressed, if it is, it checks if a printer is online, (to
870 prevent a dead loop if the key is pressed accidentally with
880 no printer attached). Both of these conditions being true,
890 the routine then sends to the printer the contents of the
900 80 col screen (either 23 or 24 lines, alter PAGE to suit).
910 At any time, a CTRL STOP will stop the printout, but, as
920 this would put an I/O ERROR on the screen (and might not
930 allow the CP/M version to return to CP/M) a CTRL C will
940 also stop the printout (by jumping to the exit) and this
950 also maintains compatabilty with the CP/M dos.

960
970 As the routine has been written with CP/M in mind, it is
980 fully relocatable, apart from the ALTRHk routine at the
990 start. Bear in mind that at the time this routine is called,
1000 every 1/30 second, the BASIC rom is in lower memory,
1010 regardless of whether CP/M or BASIC is running.

1020
1030 The printer must be set up to print 80 chars per line, i.e
1040 not condensed mode. Emphasized, bold and double strike are
1050 all ok though.

1060
1070 Graphic characters will, with this version, produce
1080 unpredictable results. After I've converted this one to
1090 run under CP/M, I'll have a go at version 1.1 to correctly
1100 print the SVI graphic characters, this will be a lot more
1110 machine (printer) specific though. Mine is a BRANDT CP 80
1120 which is compatable with the EPSON 'standard'.
1130 END

PROGRAM TO PROGRAM DATA TRANSFERS

=====

By Bryan Fisk

=====

The following two programs "mainprog" and "sub1" illustrate a method of transferring data from one program to another. The method employed here is to store any data that is to be transferred between two or more programs in a single array - this makes the saving/loading of the data into a disk file much easier to program/maintain than using many variables and saving each individual variable.

The disadvantage with this method is that instead of having 'meaningful' variable names the program must use array items eg. 'input "number of items",v(1)' instead of 'input "number of items",numitems'

To make the program easier to debug (and to document what the array items represent), a second array is employed - each item of which consists of a description of the variable array item eg.

v(1) ---> v\$(1) ---> "start_number"

v(2) ---> v\$(2) ---> "temp_count"

Below is

Below is the program flow, between "mainprog" and "sub1"

mainprog

=====

- a check is done to see if a data file ("1:main.dat") exists [10 - 30]
- finds there isn't a file , so initiates the variable data/name tables [120 - 220]
- v(1) (start_number) is set to 1 [1005]
- v(3) (subroutine_flag) is checked and found to be 0 [1010]
- the numbers 1,2,3,4,5 are printed [2000 - 2060]
- the variable data/name tables are saved as "1:main.dat" [4000-4070]
- the file "1:sub1" is run [2070]

sub1

=====

- a check is done to see if a data file ("1:main.dat") exists [10 - 30]
- loads the variable tables from it [40 - 110]
- the numbers 5,4,3,2,1 are printed [1000 - 1050]
- v(3) (subroutine_flag) is set to 1 [1070]
- the variable table is saved as "1:main.dat" [4000 - 4070]
- the file "1:mainprog" is run [1090]

mainprog

=====

- a check is done to see if a data file ("1:main.dat") exists [10 - 30]
- finds there is a file , so loads the variable tables from it [40 - 110]
- v(1) (start_number) is set to 1 [1005]
- v(3) (subroutine_flag) is checked and found to be 1 [1010]
- the numbers 1,3,5 are printed [3000 - 3060]
- the variable data/name tables is printed [6100 - 6160]
- the data file ("1:main.data") is deleted [1025]
- mainprog ends [1030]

PROGRAM TO PROGRAM DATA TRANSFERS (cont)

=====

```

1 ' main program
2 '=====
3 '
10 ' read variable table if one exists
-- sts
20 ON ERROR GOTO 6000
30 OPEN "1:main.dat" FOR INPUT AS #
-- 1
40 ON ERROR GOTO 0
50 INPUT #1, A
60 DIM V$(A), V(A)
70 FOR X=1 TO A
80     INPUT #1, V$(X), V(X)
90 NEXT X
100 CLOSE #1
110 GOTO 1000 ' main program
111 '
112 '
113 '
120 ' initialize variable table
130 ON ERROR GOTO 0
140 READ A
150 DIM V$(A), V(A)
160 FOR X=1 TO A
170     READ V$(X)
180 NEXT X
190 DATA 4
200 DATA "start_number", "temp_cou
-- nt"
210 DATA "subroutine_flag", "max_v
-- ar_name_length"
220 GOTO 1000 ' main program
230 '
240 '
250 '
1000 ' main program
1005 V(1)=1
1010 IF V(3)=0 THEN GOSUB 2000 ELSE
-- GOSUB 3000
1020 GOSUB 6100 ' print variable da
-- ta/names
1025 KILL "1:main.dat"
1030 END
1040 '
1050 '
2000 ' print start_number to 5
2005 PRINT "main program"
2010 V(2)=V(1)
2020 IF V(2)>5 THEN 2060
2030 PRINT V(2)
2040 V(2)=V(2)+1
2050 GOTO 2020
2055 '
2060 GOSUB 4000 ' save variable tab
-- le
2070 RUN "1:sub1" ' run subroutine
-- 1
2080 '
2090 '
2100 '
3000 ' print start_number to 5 step
-- ping2
3005 PRINT "main program"
3010 V(2)=V(1)
3020 IF V(2)>5 THEN 3060
3030 PRINT V(2)
3040 V(2)=V(2)+2
3050 GOTO 3020
3055 '
3060 RETURN
3070 '
3080 '
3090 '
4000 ' save variable table
4010 OPEN "1:main.dat" FOR OUTPUT A
-- S #1
4020 PRINT #1, A;
4030 FOR X=1 TO A
4040     PRINT #1, V$(X);",";V(X);
4050 NEXT X
4060 CLOSE #1
4070 RETURN
4080 '
4090 '
4100 '
6000 ' initialize variable table as
-- data file does not exist
6010 RESUME 120
6020 '
6030 '
6040 '
6100 ' print all variable names & v
-- alues
6110 FOR X=1 TO A
6120     IF V(4)<LEN(V$(X)) THEN V(
-- 4)=LEN(V$(X))
6130 NEXT X
6135 '
6140 FOR X=1 TO A
6150     PRINT V$(X);SPACE$(V(4)-LE
-- N(V$(X))); " = V(";X;") =";V(X)
6160 NEXT X
6170 RETURN

```

PROGRAM TO PROGRAM DATA TRANSFERS (cont)

```

1 ' subroutine 1
2 '=====
3 '
10 ' read variable table if one exists
-- sts
20 ON ERROR GOTO 6000
30 OPEN "1:main.dat" FOR INPUT AS #1
-- 1
40 ON ERROR GOTO 0
50 INPUT #1, A
60 DIM V$(A), V(A)
70 FOR X=1 TO A
80 INPUT #1, V$(X), V(X)
90 NEXT X
100 CLOSE #1
110 GOTO 1000 ' main program
111 '
112 '
113 '
1000 'print 5 down to start_number
1005 PRINT "subroutine 1"
1010 V(2)=5
1020 IF V(2)<V(1) THEN 1070
1030 PRINT V(2)
1040 V(2)=V(2)-1
1050 GOTO 1020
1060 '
1070 V(3)=1
1080 GOSUB 4000 ' save variable table
-- le
1090 RUN "1:mainprog" ' return to main
-- in program
1100 '
1110 '
1120 '
4000 ' save variable table
4010 OPEN "1:main.dat" FOR OUTPUT AS #1
-- S #1
4020 PRINT #1, A;
4030 FOR X=1 TO A
4040 PRINT #1, V$(X);",",V(X);
4050 NEXT X
4060 CLOSE #1
4070 RETURN
4080 '
4090 '
4100 '
6000 ' initialize variable table as
-- data file does not exist
6010 RESUME 120
6020 '
6030 '
6040 '
6100 ' print all variable names & values
-- alues
6110 FOR X=1 TO A
6120 IF V(4)<LEN(V$(X)) THEN V(4)=LEN(V$(X))
6130 NEXT X
6135 '
6140 FOR X=1 TO A
6150 PRINT V$(X);SPACE$(V(4)-LEN(V$(X)));",",V(X)
6160 NEXT X

```

***** SPECTRAVIDEO DISC MANAGER BY Jim Oliver

There are now many excellent disc managers around. We have NSWEEP.BAS from the West German group, Bryan Fisk's DISK.UTL and SVFD from Elliot Young. However all of these have one thing in common and that is they are all large files requiring large overheads in terms of disc space and memory requirements. Also they have many features which are not used often. All of these programs run rather slowly due to the large size of the file.

What I felt I needed was something much smaller which I could keep on nearly all my discs and which would do the following :

LOAD FILES : RENAME FILES : DELETE FILES : VIEW FILES : COPY FILES

I made one previous attempt at doing this with SWP.30. I now find that I use other disc managers in place of this.

I have therefore designed this new one to run directly from the

FILES printout on the screen, which I know some of you do now anyway. This new disc manager should make things easier and faster for you.

Firstly type in and save this program on your disc as "SVDM". If this disc has system tracks then type IPL "RUN"+CHR\$(34)+"1:SVDM" (with the disc in drive 1) and then hit return. This then tells the computer to load and run the program called "SVDM" directly after it boots up.

After the computer boots (starts up) you will see a list of the programs or files on the disc presented on the screen and a small "<" character alongside the top righthand file name. The function keys will have changed and will now let you operate on any of the programs as you wish by altering the position of the pointer "<" (or ">" by pushing the 'cursor left' key) by using the cursor keys. Do not try to run the cursor off the top of the screen or an error will result ! I haven't mug-trapped EVERY problem, you still have to think a little !

If you want to change to the other drive just hit "1" or "2". To Quit just hit "Q" (or "q").

The file size is about 1900 bytes in memory but can be reduced to 1700 by removing all the spaces and REM's. Careful with the REM line numbers though !

The program works by PEEKing the video ram and using this information to operate on the file according to your fancy. Line number 560 contains the function key definitions I use, but you may enter your favourites here if you wish.

For the experts in the group : please MODIFY IT --- IMPROVE IT --- then let me have it back so I can share the improvements with others.

```

10 REM SV DISC MANAGER          -- is ASCII file ";R$:IF R$="R" OR
20 MAXFILES=2: CLEAR 1000: SCREEN 0,1 -- R$="r" THEN A=1:GOSUB 550:LOAD F
30 IV$=CHR$(27)+"p":NV$=CHR$(27)+"q  -- $,R ELSE 380
-- ":WIDTH 40                  300 RETURN 110
40 ON KEY GOSUB 240,310,340,370,440 310 REM RENAME
50 CLS:REM J.A.OLIVER WSUG MAY 88    320 GOSUB 510:PRINT:PRINT" New fi
60 PRINTIV$;" Which drive to use "; -- le name ";F$:STRING$(15,29);:I
-- NV$;:INPUT D$:IF VAL(D$)<1 OR VA -- NPUT NF$:NAME F$ AS NF$:FOR T=3
-- L(D$)>2 THEN BEEP:GOTO 60        -- TO LEN(NF$):VPOKE (X1-3)+T,ASC(M
-- ID$(NF$,T,1))-32:NEXT:RETURN 110
70 CLS                            330 RETURN 110
80 CLOSE:FILES VAL(D$):GOSUB 480    340 REM DELETE
90 P=28:X=12:Y=1:B=46:MC=42:SS=35:C 350 GOSUB 510:PRINT:PRINT"Delete";I
-- S=0                          -- V$;" y/n "NV$;:INPUT DL$:
100 CS=CS+40:IF VPEEK(CS)<>0 THEN 1 -- IF DL$="n" OR DL$="N" THEN RETURN
-- 00                            -- 110 ELSE KILL F$:FOR T=X1 TO X1+12:
--                               -- VPOKE T,106:NEXT
--                               360 RETURN 110
--                               370 GOSUB 510
110 F$="":DV$="CRT":A=0:LOCATE 0,C   380 PRINT:INPUT "(V)iew or (P)rint
-- S/40:PRINT CHR$(27)+"J";        -- out";P$:IF P$="P" OR P$="p" THEN
120 VPOKE X,P                      -- DV$="LPT:"
130 A$=INKEY$:IF A$="" THEN 130     390 OPEN DV$ FOR OUTPUT AS #1:OPEN

```

```

140 IF A$="Q" OR A$="q" THEN GOSUB -- F$ FOR INPUT AS #2
-- 550 400 IF EOF(2) THEN CLOSE:RETURN 50
150 IF A$=CHR$(31) THEN GOSUB 220:X= 410 LINE INPUT #2,L$:PRINT #1,L$
-- X+40:GOSUB 230 420 GOTO 400
160 IF A$=CHR$(30) THEN GOSUB 220:X= 430 RETURN 110
-- X-40:GOSUB 230 440 REM COPY
170 IF A$=CHR$(28) THEN GOSUB 220:X= 450 GOSUB 520
-- 28:GOSUB 230 460 D=VAL(D$):PRINT:PRINT"COPYING "
180 IF A$=CHR$(29) THEN GOSUB 220:X= -- ;F$;"TO DRIVE ";CINT(3-D):COPY
-- CINT(3-D) FROM F$
-- 12:GOSUB 230 470 RETURN 110
190 IF A$="2" THEN D$="2":GOTO 70 480 RESTORE 490:FOR T=1 TO 5:READ A
200 IF A$="1" THEN D$="1":GOTO 70 -- $:KEY T,A$:KEY (T)ON:NEXT T
210 GOTO 130 490 DATA " LOAD"," RENAME"," DELETE
220 VPOKE X,0:RETURN -- "," VIEW"," COPY"
230 VPOKE X,P:RETURN 500 RETURN
240 REM LOAD 510 REM get file name and join driv
250 GOSUB 510 -- e
260 IF FA=B THEN A=1:GOSUB 550:LOA 520 X1=X-12:REM SET POINTER
-- D F$:RETURN110 530 FOR T=X1 TO X1+10:F$=F$+CHR$(VP
270 IF FA=SS THEN A=1:GOSUB 550:LO -- EEK(T)+32):NEXT:FA=ASC(MID$(F$,
-- AD F$:RETURN60 7,1)):MID$(F$,7,1)="." :F$=D$+"." +F$:RETURN
280 IF FA=MC THEN A=1:GOSUB 550:BL 540 REM RESET F/KEYS
-- OAD F$,R:RETURN 110 550 FOR T=1 TO 4:KEY(T)OFF:NEXT
290 PRINT:INPUT" (R)un or (V)iew th 560 KEY1," close:files ":KEY 2," load
-- " +CHR$(34)+"1":KEY 3," save " +CHR$(34)+
-- "1":KEY 4," list":KEY5," run" +CHR$(13)
-- :KEY 6," color15,4,5" +CHR$(13):
-- KEY 7," load "
-- +CHR$(34)+"2":KEY 8," save " +CHR$
-- (34)+"2:"
570 IF A THEN RETURN ELSE END

```

REVIEW

ConIX operating system

... "A powerfull user-interface enhancement"...

... Bob Curwood

Well, thats what the blurb says, lets see if it lives up to it !

ConIX is a front-end replacement for the CCP which is the part of CP/M which accepts your commands and requests for files, programs etc and then processes them (if it can!) As such, ConIX is very visible to the user and has lots of 'up-front' goodies to keep you hooked on using it.

There is, of course, a price to pay for these and it shows in two ways, a slight slowing down in response when running a program (such as Wordstar which I'm using for this review) and a drastic reduction in the size of the TPA, which is the amount of free memory available to run programs.

The first of these is no problem and the second is cleverly bypassed by a system of memory management. Conix can be split into several segments and the level necessary for any particular program to run can be set in a number of ways. The minimum amount

of space is only 1/2k but this is only a loader which will reload Conix after your program is run. The maximum is about 25K but as a lot of CP/M programs/utilities are quite small they will run in the complete Conix environment.

Ok, some 'pro & cons', all mixed together, to try and give you a feel for the system and whether it will work for you.

User areas I doubt if most SVI cpm users ever bother with the user command, once logged into another area (0 is the default with 1 to 15 also available) you dont have any files or programs available except those you have stored on disk in that area. You cant (except with programs like NSWP) access programs in other areas. Conix has 32 user areas (you always know which one you are in because of the prompt <A:00> to <A:31>) and these can be used to store different types of programs on the one disk (eg utilities, data etc) and they can all be accessed from whatever area you are in.

Redirection This is the feature of Conix I really like. You can have upto 16 drive/user combinations (such as A:00 B:00 A:01 etc) and when you request a program Conix will search through these to find it. A similar feature is used to allow programs like Wordstar, which expects to find it's overlays on a particular drive, to be used from whatever drive/area you happen to be in.

Redirection is much more than this tho', all inputs (from console, files, programs etc) and outputs can be redirected to other devices, files etc, either instead of, or in addition to, the normal expectations. Files can also be 'piped', i.e the output of one file becomes the input of another. This allows virtually automatic operation of any program as it no longer needs you at the keyboard to enter responses, they can come from a file (or the RS232 port perhaps)

Disk Full! Ever had that error? Very annoying when you have just finished a long letter or source code and find you have no room to save it on disk. Even worse if both drives are full! Conix has a feature called ExpanDisk which lets you then save it to a logical drive(s) with the defaults of I:,J: or K: on the physical drive A. You are prompted to remove the current disk, load in the disk to save on then replace the current disk. All the defaults can of course be changed, as can the drive to be used.

DIR,ERA,REN,SAVE,TYPE & USER These are the only 'inbuilt' commands of CP/M and they are supplemented by a very large number of utility commands in Conix, 39 of them in fact, (one of which, DPT has about 36+ arguments). A quick list of the interesting ones are ATRIB to set file attributes, BDEC, CHR, DECB, HXPR to convert to/from bin dec & hex, EXAM, FILL, MOVE, SCRC, ZAP, which all work on memory, IN/OUT from a port, COPY, DEL, FIND, READ, WRITE, UDIR which are for files, (the last one UDIR can list all files in all user areas) and the variables which include function keys, character keys memory variables and a set of disk based variables. (Did you notice this review has both ' and ' ? Not to mention !) Just as well, because the ! ascii character is used

in the pipe commands as well as meaning 'or' in the XCC command language! (P.S there are 24 function keys each of which will produce a single ascii character, the 52 disk variables can each contain upto 255 bytes and are saved in a disk file which is read each time a var: is referenced (direct or from a program))

Another feature which new users would find very useful is the ^B which loads and runs (by default, you CAN change it) a menu based help system which is very extensive. (There is also a MANual program which outputs from various archived manual files).

Any users who write CP/M source code programs will be interested to hear about the additional BDOS calls (MVI C,call, CALL 0005) which range from 200 and up, e.g 211: Initialize FCB. 214: Open FCB which is similar to BDOS 15 but also allows automatic file searching when the filename is preceded by a colon, when (if) the file is found, the FCB is altered to the correct drive/user ready for any subsequent I/O. There are currently 17 extra BDOS calls.

COM programs can be nested within the the Conix environment so that one COM program can call another which calls another.... This is accomplished storing information about the parent program(s) in a special file on disk, aprox 1k is needed for each level of nesting and it is possible to break out of 1 or more levels of nest or exit direct to the CCP prompt.

Language Interpreter Conix enables you to write programs in a high level language (Not as easy to understand as Basic but better than M/C!). This requires the source you write to be interpreted by XCC.COM which produces an intermediate code file which (only) Conix can execute. Unfortunately, the copy of Conix lent to me for review did not contain XCC.COM so I couldn't try this out.

Archives A major program in the Conix system is ARM.COM the archive manager. Like LU.COM this puts many small (or otherwise!) programs/files etc into one archive thus saving lots of disk space since even a 1 byte program will take 1k (2k in d/s systems) space on the disk and one of the directory slots (64). Unlike LU, Conix lets you RUN archived programs without explicitly extracting them first. Each archive can contain hundreds of small programs. If each one contained only 1 record you would save over 75% in disk space. Each file can be time/date stamped and have a short (<35 chars) description tagged to it. Programs stored in the default archive can be included in the automatic search path directory, i.e after looking for a program in A:00, B:00, A:31 etc (See first page) and it isnt found, Conix will then look in %.ARC as well. (The default archive is also used by the MANual program and can, of course, be changed).

The review conclusions

Is ConIX worth getting for YOUR system? It depends! If you use your SVI for business and you are not into writing your own programs, Conix can be of use to you, especially the file path redirection/ searching, but you might need some help to install it. If you are an average SVI user, i.e you are slowly learning about CP/M then I think you will like Conix, and will very

quickly decide to have it in memory all the time. Be warned that you will need to study the manuals well 'till you get the hang of it (but the online help system, called with a ^B is great!). If you are a programmer, you must have CONIX, it's few disadvantages are far outweighed by the many features. However, programmes written in XCC command programs can only be used by other users with Conix, (The first byte of such a program is C9 !). Since it's been out for a few years now, there might even be a 'ConIX Users Group' going in the states, I expect one will get going here.

I will be sending for it!

Note : If you are interested in sending for a copy of the ConIX system please post a stamped addressed envelope to Jim Oliver and he will send you a photo copy of the original advertisement. The approximate cost for the system as reviewed above is US \$70.00 including the ConIX programming system manual and postage by air from USA. To get the ConIX "compiler" (XCC.COM) referred to above is an extra US \$30.00.

Spectravideo in Hong Kong

Henry Chiu

I recently went on a trip to Hong Kong. While I was there I rang up "Bondwell" to find out the latest situation regarding Spectravideo computers. There were several telephone numbers and after some persistence I was able to get hold of the Spectravideo Product Manager.

Initially their Product Manager, Mr Au Yeung (I think that's his name) was not very forthcoming with information. He stated that I should contact the local New Zealand Agent Pegasus who are situated in Hamilton.

I then gave him a brief lesson in New Zealand geography and explained that I was a local dealer in Wellington of SVI and MSX computers, and since the demise of CDL it is difficult to obtain hardware and software for the above computers.

Au Yeung explained that he had only been with Bondwell/SVI for just over a month and was not too familiar with the above computers. He was however good enough to do a spot investigation for me by talking to his supervisor who had been with Bondwell a lot longer.

Firstly, the SVI 318 and 328 are dead, they have long ceased production on these products, (This we all know).

Secondly, SVI are still producing MSX computers and exporting to S.E. Asia but he was unable (or should I say unwilling) to disclose the exact destination or in fact whether they were sold as SVI branded computers.

Thirdly, MSX never took off in Hong Kong so SVI have concentrated their marketing on their range of IBM clones, joysticks and computer toys such as laser guns and 2nd generation TV games.

Lastly I asked whether they were in a position to supply spare parts/hardware for the 318/328 and MSX computers. Au Leung's advice to me was to collate all our orders together then fax them with the details and if a production run was viable then they would certainly consider fulfilling our orders. The main thing is that we get enough orders together, so perhaps those interested could write to the club and state your requirements.

It was also interesting to note that on speaking to dealers in Hong Kong they preferred not to stock the Bondwell/SVI range of computers as they were too expensive and did not offer the expansion capabilities offered by some of the Taiwanese and Korean IBM clones. Also Bondwell/SVI have closed up their "New World Office Building" retail outlet in Tsim Sha Tsui, the prime tourist area in Hong Kong, so apart from toys and joysticks they appear to have lost their market presence domestically. This being a complete reversal from 3 years ago where virtually every Camera, Toy and Hobby shop in Hong Kong were stocking SVI products.

The Hong Kong Scene

The market has been divided into two camps:-

a. 2nd Generation computer games machines e.g. Sega and various Hong Kong made copies retailing at around NZ\$150.
(rom cartridges for the games cost around NZ\$15 ea.)

b. IBM and the many clones.

It was refreshing not to hear the names Commodore or Amstrad bandied around. In fact the clones are so reasonably priced it is not worth considering anything else. An IBM XT clone with 20 Meg hard disc, twin floppys, green monitor and quality parts would set you back NZ\$1000, (Quality meaning Japanese and not Korean parts). Also it is taken for granted that you get a 640k ram with a least 6 expansion slots.

I spent half a day at the "Golden Shopping Centre" at Sham Sui Po. This is really an incredible place where approximately 200 computer shops are housed together. There were stores that specialised in hardware, others in pirated software and literature. Just to give an idea on prices, software such as Lotus and Dbase could be bought for the price of the disc. Accompanying literature would cost roughly NZ\$15. It was amusing to see that the copyright warnings were also printed in the pirated books and some even had registration cards to send back to Ashton Tate etc. Whether anyone would send back the card is another matter.

If you were really adventurous you could buy all your bits and

pieces from individual shops to make up your computer. Prices varied so much between shops for various components. Some shops stocked under the counter a direct copy of the IBM bios and if you bought the bios and cabinet housing the CPU they would attach the IBM markings with it.

On walking through the "Golden Shopping Centre" apart from the IBM compatible joysticks, only one shop bore any reference to Spectravideo and namely MSX, but unfortunately this shop was shut. The neighbouring shop said they were shut permanently, so this again emphasizes the lack of presence of SVI.

From the situation in Hong Kong it has become more and more apparent that the days of the 8 bit computers when used as a business tool are numbered. Even for a computer as universally popular as the Apple II's series it was difficult to get software for. Talking from a commercial point of view where new and more sophisticated software has come available it is quite apparent that the more powerful clones have dominated the market. However when I spoke to one dealer who was more an enthusiast than a businessman he still seemed to think that there is still a lot going for the 8 bit machine especially if it is a Z80 and can run CP/M. We all know that you don't need an MS-Dos machine to learn and run Machine Code, Assembly, Pascal or other languages.

The dealer was quite honest in saying that the great majority of first time buyers did not know what they wanted. Usually they bought the computer with the largest memory then after realising what they have bought have ended up having to buy one of these second generation T.V. games machines for their children. Also the dealer showed me an article in a recent U.S. computer magazine which stated that there has been a resurgence of interest in CP/M and a lot of new programs have been written recently by professional programmers [did you get a copy?...ed].

All in all the Hong Kong experience was quite worthwhile and it again demonstrates that our own 318/328 machines still have a lot to offer even though the technology is a little out of date.

NEXT ISSUE

A BEAUT ARTIST PROGRAM IN BASIC ACCESSING DISK/TAPE SAVED IMAGES

WORDPROCESSING IN OTHER LANGUAGES

WHAT'S WRONG WITH DATASTAR?

MEMBERSHIP LIST

plus! plus! plus!

YOUR CONTRIBUTION

DO YOU REALLY NEED A CLONE?

Many users of 8-bit computers have been asking themselves whether they should "upgrade" to an IBM PC-compatible computer ("Clone" for short). I am writing this as one who owns both an SVI 328 and a Magnum Mk II XT-20 (Clone).

First of all why would anyone contemplate such a change? Higher speed? The SVIs run at 3.6 MHz; the Clones at 4.77 MHz. Not much faster unless you have a turbo model at say 8.0 MHz. Sixteen bits instead of eight? The clones aren't true 16-bit computers anyway. Extra memory? Sure: 640 Kbytes on mine. Fancy graphics? You've got to be kidding.

Let's start with that point. The graphics on the original IBM PCs were an optional extra, and so poor that many other companies brought out improved plug-in graphics cards. Consequently there is no real standard graphics display on clones. The original IBM colour graphics adapter (CGA) could display only four colours at a time out of a palette of 16. Definitely not as good as on the SVI computers. Although the pixel resolution is higher (600x200) it is only significantly so in the horizontal direction (cp. SVI's 256x192). Since then other displays have appeared, eg. the enhanced graphics adapter (EGA), the Hercules graphics card (HGC), which has 720x348 pixels, and recently the HGC Plus, which I have and it gives me many of the graphics facilities of the SVI and multiple user defined character fonts, but only in monochrome and no sprites! There is also the Hercules Incolor card, and there are many others. All these offer different facilities in the way of colours and resolution. Consequently programs which run on one graphics output wont necessarily run satisfactorily on another. Recently multi-function cards have appeared which can be switched between one mode and another, but they are expensive.

How about sound? The SVI sound chip has three channels and so can play two or three part harmony. Some of the games (such as "Telebunny") have quite impressive music. The IBM on the other hand has only one channel of sound, so no harmony.

The versions of BASIC available on the clones are quite similar to that on the SVI, but with no graphics, unless you have the right graphics card, and NO SPRITES!. Actually, I never use BASIC, on the Clone, because I prefer to program in Turbo Pascal. Turbo Pascal is of course available for CP/M machines, and I have it on the SVI, so many of my programs have been transferred to the Clone and simply recompiled. Graphics programs have to be partly rewritten. Modula-2 is available for both CP/M and MSDOS machines as are many of the older programming languages.

How about MSDOS, the operating system used on the Clones? MSDOS is actually very similar to CP/M and was obviously based on it. Many of the commands are the same or similar. There are extra ones, but they are mostly to do with hard disks and subdirectories. When the IBM displays the directory it includes the date and time at which each file was created. This can be very useful when cataloging files. In case you didn't know, you could have such extra facilities and more on your SVI. All you need to do is to buy one of the new swept-up versions of CP/M. First there was ZCPR3. Now there is Z-system, which includes ZCPR3 and ZRDOS, and is faster than CP/M. QP/M gives you automatic disk relogging (so you don't

have to press Ctrl C whenever you change disks), time and date stamping, etc and all in the same amount of memory as your CP/M 2.2 takes. "Conix" advertises that their new replacement for CP/M has 100 new commands, including print spooling (so you can go on using your computer while it is sending things to the printer), screen paging, archiving, I/O redirection etc. Pretty good for \$US30. An update to MSDOS 3.2 costs about \$NZ150.

That's another thing: much of the commercial software for Clones is more expensive than for 8-bit computers.

By now you may be wondering why anyone in their right mind would buy a Clone. I bought a Clone because my SVI-328 suffered multiple chip failure when the power supply blew (probably because my Mark 1 came unstuck in the middle). It took months to get it fixed, (some of the chips had to be imported from the UK) and it seemed that it might never make it.

Okay, the Clone does have some advantages. First and foremost it has a 20 megabyte hard disk. This certainly speeds up working on it, especially programming. It obviates most of the need for changing floppy disks, except to make backup copies of everything. But you could have a hard disk on your SVI too.

Secondly it gives me access to an ever growing mountain of available software, most of which I can't afford. Fortunately there is a lot a public domain, free-ware and share-ware software around. Much more than anyone would ever need, so it takes a while to work out what is worth having and what isn't.

Apart from that there aren't many real advantages in having a Clone, so if I hadn't been forced into it by the apparent demise of the SVI-328 I wouldn't have bought one. I could have waited for the price of the next generation to come down and then made a real advance.

Donald Rogers

LAST NOTICE

THIS EDITION HAS BEEN SENT TO YOU EVEN IF YOU EXPIRED AS LONG AGO AS JANUARY 1988. PLEASE CHECK YOUR EXPIRY DATE ON THE FRONT LABEL AND SEND YOUR \$18 IN TODAY!

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